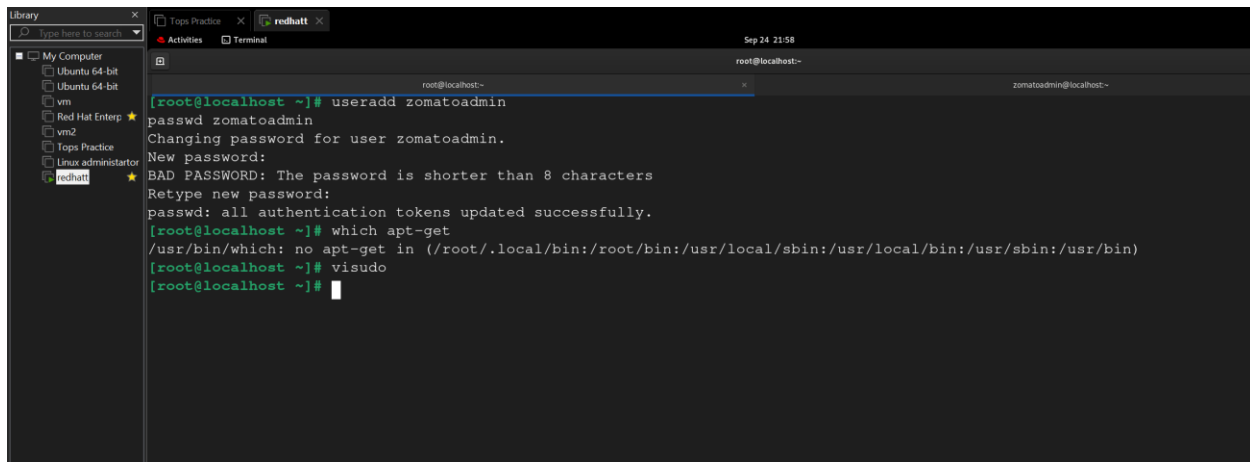


1.

Create a new user account called zomatoadmin on your Linux system, and configure it so that this user can run only the apt-get update and apt-get upgrade commands using sudo without entering a password.

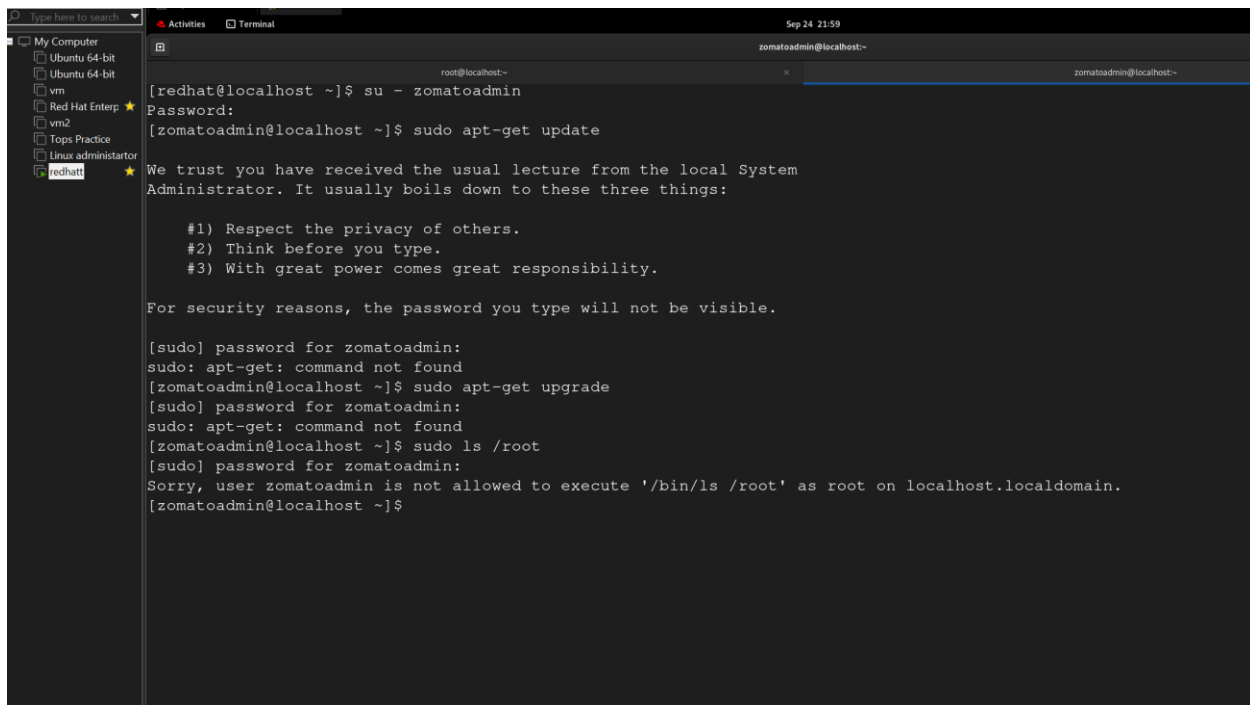
Hint: Edit the /etc/sudoers file using visudo and use the NOPASSWD directive for specific commands.

>



A terminal window on a Linux system. The user is root at localhost. The commands executed are: `useradd zomatoadmin`, `passwd zomatoadmin`, and `which apt-get`. The password for zomatoadmin is set, and the output of `which apt-get` is shown. The terminal also shows the output of `visudo`.

```
[root@localhost ~]# useradd zomatoadmin
passwd zomatoadmin
Changing password for user zomatoadmin.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# which apt-get
/usr/bin/which: no apt-get in (/root/.local/bin:/root/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin)
[root@localhost ~]# visudo
[root@localhost ~]#
```



A terminal window showing the configuration of sudo privileges for the zomatoadmin user. The user switches to zomatoadmin using `su - zomatoadmin`. The user then runs `sudo apt-get update` and `sudo apt-get upgrade`, which succeed without a password. The user then runs `sudo ls /root`, which fails with a permission error. The terminal also shows the output of `visudo`.

```
[redhat@localhost ~]$ su - zomatoadmin
Password:
[zomatoadmin@localhost ~]$ sudo apt-get update
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for zomatoadmin:
sudo: apt-get: command not found
[zomatoadmin@localhost ~]$ sudo apt-get upgrade
[sudo] password for zomatoadmin:
sudo: apt-get: command not found
[zomatoadmin@localhost ~]$ sudo ls /root
[sudo] password for zomatoadmin:
Sorry, user zomatoadmin is not allowed to execute '/bin/ls /root' as root on localhost.localdomain.
[zomatoadmin@localhost ~]$
```

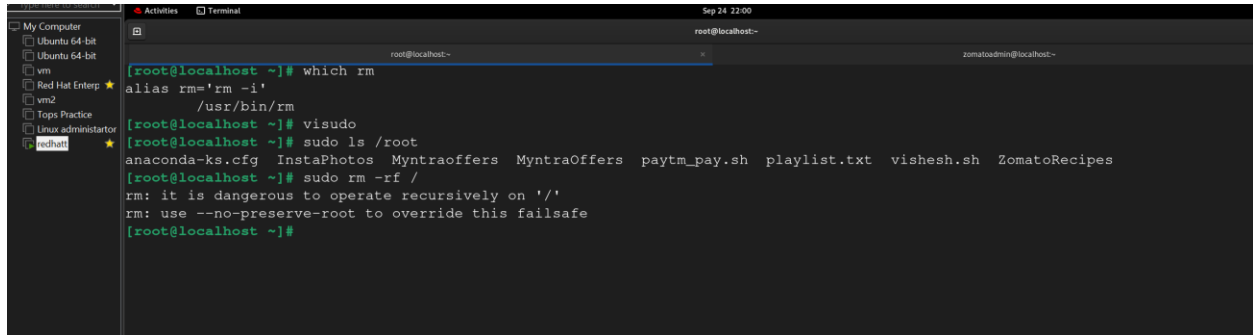
2.

Modify the sudo privileges for your existing user so that they cannot run `sudo rm -rf /` but can run all other sudo commands.

Constraint: Use

command-specific restrictions in the sudoers file to block only the dangerous command.

>

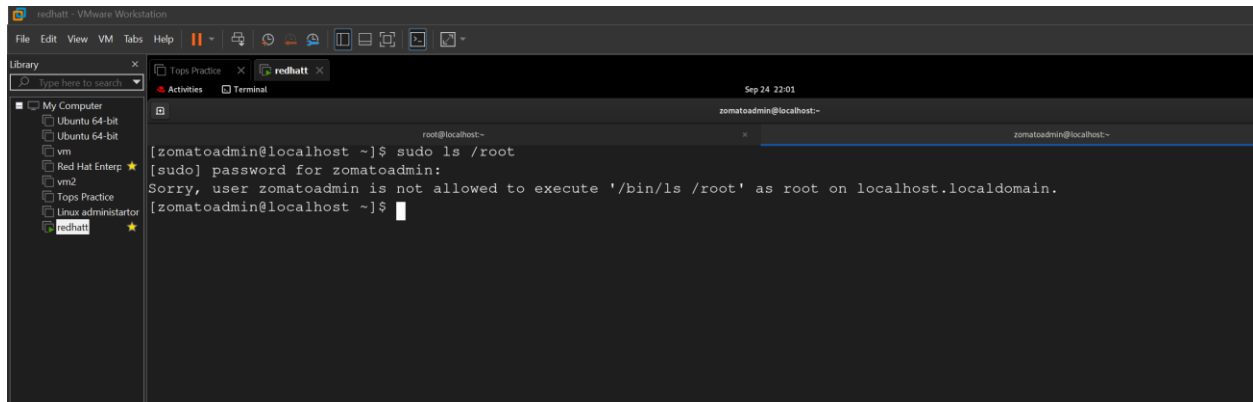


```
[root@localhost ~]# which rm
/usr/bin/rm
[root@localhost ~]# alias rm='rm -i'
alias rm='rm -i'
[root@localhost ~]# visudo
[root@localhost ~]# sudo ls /root
anaconda-ks.cfg  InstaPhotos  MyntraOffers  MyntraOffers  paytm_pay.sh  playlist.txt  vishesh.sh  ZomatoRecipes
[root@localhost ~]# sudo rm -rf /
rm: it is dangerous to operate recursively on '/'
rm: use --no-preserve-root to override this failsafe
[root@localhost ~]#
```

3.

Simulate a scenario where a user tries to run a command with sudo that they are not allowed to. Take a screenshot of the error message and explain in 2-3 lines why the error occurs based on your sudoers configuration.

>

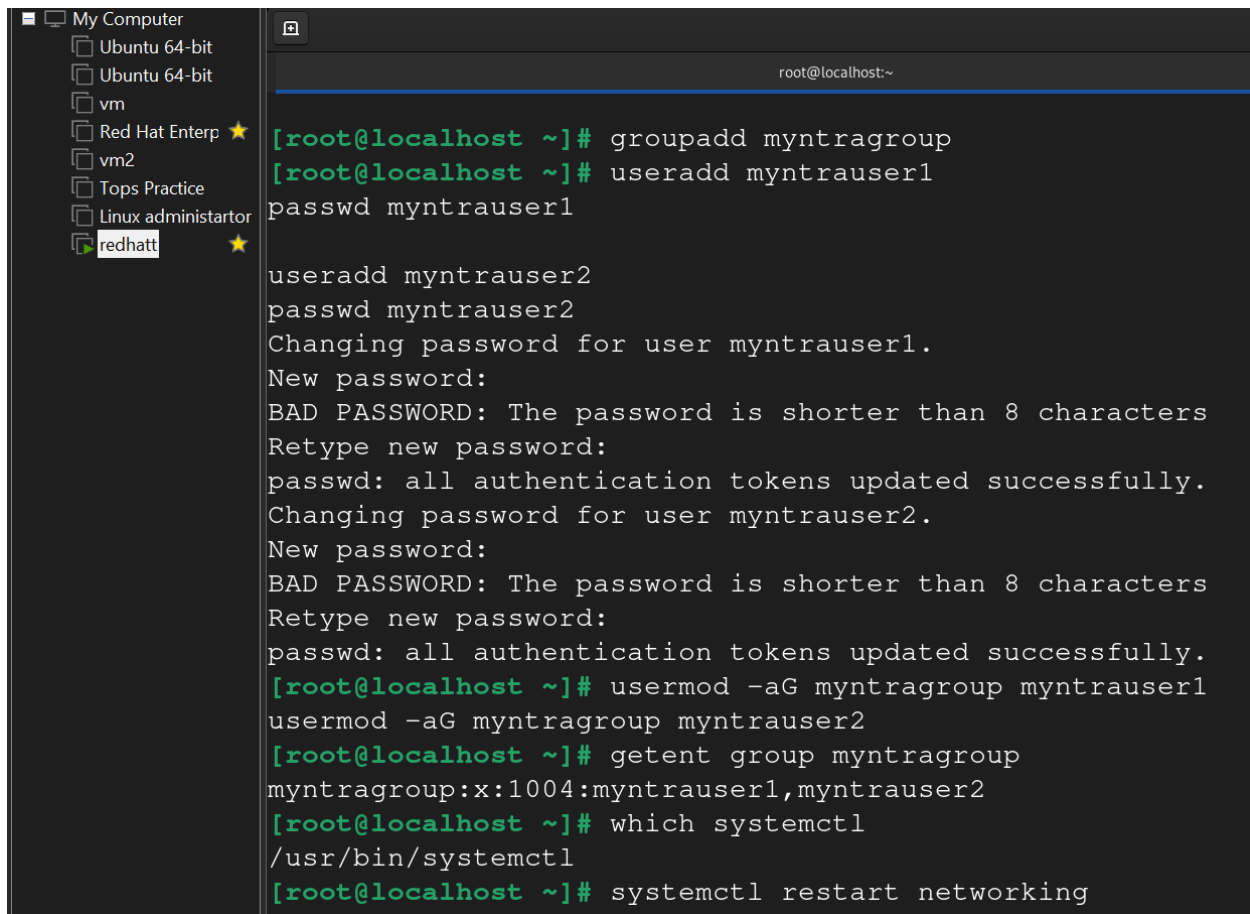


```
[zomatoadmin@localhost ~]$ sudo ls /root
[sudo] password for zomatoadmin:
Sorry, user zomatoadmin is not allowed to execute '/bin/ls /root' as root on localhost.localdomain.
[zomatoadmin@localhost ~]$
```

4.

Set up a group called myntragroup and add two users to it. Grant sudo privileges to this group so that all members can restart the networking service (systemctl restart networking) but cannot edit any files in /etc.

>



The image shows a terminal window with a sidebar on the left containing a file explorer view. The sidebar lists 'My Computer' with sub-items: 'Ubuntu 64-bit', 'Ubuntu 64-bit', 'vm', 'Red Hat Enterp' (marked with a star), 'vm2', 'Tops Practice', 'Linux administartor', and 'redhatt' (marked with a star). The terminal window title is 'root@localhost:~'. The terminal output shows the following commands and their results:

```
[root@localhost ~]# groupadd myntragroup
[root@localhost ~]# useradd myntrauser1
passwd myntrauser1
useradd myntrauser2
passwd myntrauser2
Changing password for user myntrauser1.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
Changing password for user myntrauser2.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# usermod -aG myntragroup myntrauser1
usermod -aG myntragroup myntrauser2
[root@localhost ~]# getent group myntragroup
myntragroup:x:1004:myntrauser1,myntrauser2
[root@localhost ~]# which systemctl
/usr/bin/systemctl
[root@localhost ~]# systemctl restart networking
```

```

[redhat@localhost ~]$ su - myntrauser1
Password:
[myntrauser1@localhost ~]$ sudo systemctl restart networking

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

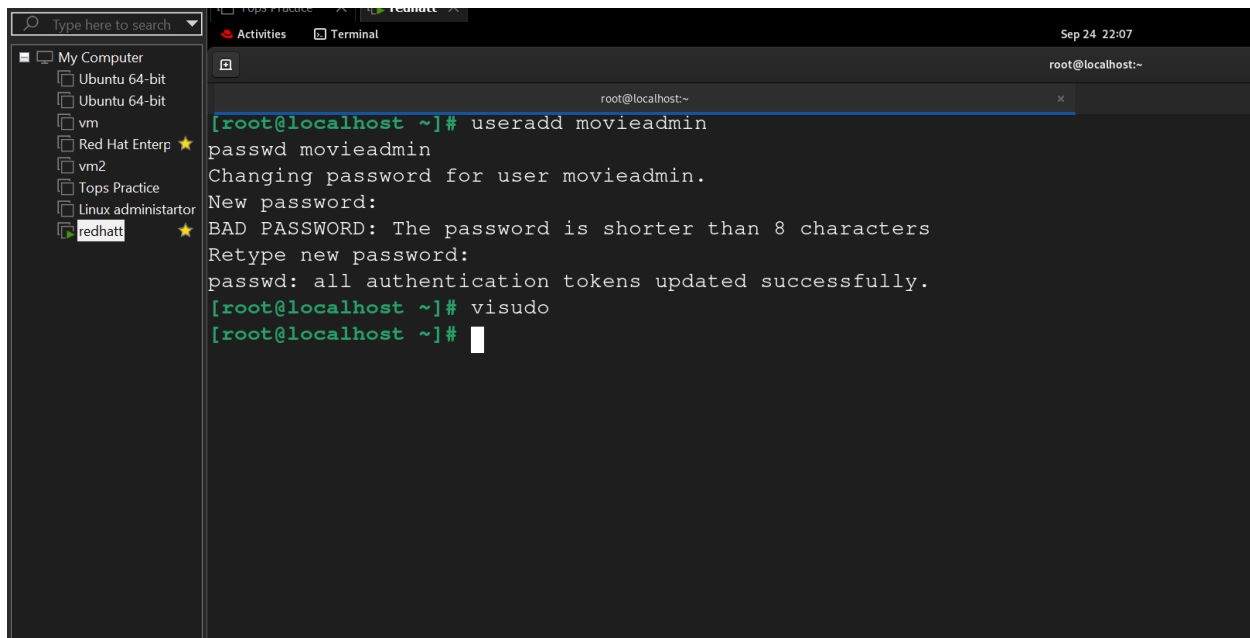
For security reasons, the password you type will not be visible.

[sudo] password for myntrauser1:
myntrauser1 is not in the sudoers file.
[myntrauser1@localhost ~]$
```

5.

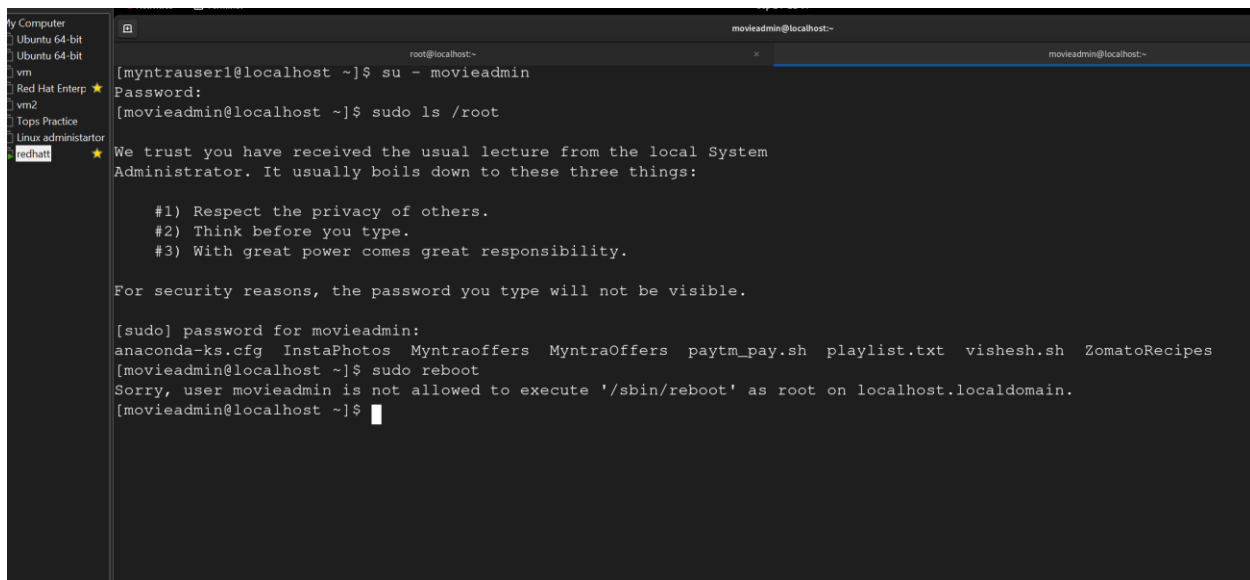
Use ChatGPT to generate a sample `/etc/sudoers` entry that allows a user called `movieadmin` to run all commands as root except for shutting down or rebooting the system. Copy the AI's suggestion, test it on your system, and write one line about whether it worked as expected.

>



The terminal window shows a root user at localhost. The user 'movieadmin' is created using 'useradd'. A password is set using 'passwd', which prompts for a new password and its confirmation. The password is rejected as being too short (less than 8 characters). After re-entering a valid password, the system message 'passwd: all authentication tokens updated successfully.' is displayed. The user then runs 'visudo' and returns to the root prompt.

```
[root@localhost ~]# useradd movieadmin
passwd movieadmin
Changing password for user movieadmin.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# visudo
[root@localhost ~]#
```



The terminal window shows a user switch from root to 'movieadmin' using 'su -'. The user then runs 'sudo ls /root', which triggers the standard sudo security warning. After entering the password, the command is executed successfully, listing the contents of the /root directory. Finally, the user attempts 'sudo reboot', which is denied with the message 'Sorry, user movieadmin is not allowed to execute '/sbin/reboot' as root on localhost.localdomain.'

```
[myntrauser1@localhost ~]$ su - movieadmin
Password:
[movieadmin@localhost ~]$ sudo ls /root
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for movieadmin:
anaconda-ks.cfg  InstaPhotos  Myntraoffers  MyntraOffers  paytm_pay.sh  playlist.txt  vishesh.sh  ZomatoRecipes
[movieadmin@localhost ~]$ sudo reboot
Sorry, user movieadmin is not allowed to execute '/sbin/reboot' as root on localhost.localdomain.
[movieadmin@localhost ~]$
```